

Institute of Actuaries of India

Subject CT3 – Probability & Mathematical Statistics

October 2015 Examination

Indicative Solution

Introduction:

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other approaches leading to a valid answer and examiners have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

(i)

Let the missing frequencies for no. of claims 1 and 3 are f_2 and f_4 respectively.

$$\text{Hence: } 75 + f_2 + f_4 = 125 \text{ giving } f_2 + f_4 = 50 \quad (1)$$

Also

$$[0(10) + 1(f_2) + 2(35) + 3(f_4) + 4(15) + 5(7) + 6(8)] / 125 = 2.504$$

$$\text{This gives } f_2 + 3(f_4) = 100 \quad (2)$$

$$(2) - (1) \text{ gives } 2f_4 = 50$$

$$\text{So } f_4 = 25$$

$$\text{From (1), } f_2 + f_4 = 50 \text{ giving } f_2 = 25 \quad [3]$$

(ii)

The median is equal to the 63rd $((125 + 1) / 2)$ observation; which is 2.

The mean is 2.504. As mean > median the data is positively skewed. [2]

[5 Marks]**Solution 2:**

(i)

Given that the mean of the binomial distribution is $(np) = 12$ and $n = 20$.

Hence $p = 0.6$. That is the distribution is binomial $(20, 0.6)$

The PGF of binomial distribution is given by

$$\begin{aligned} G_X(t) &= E(t^X) = \sum_{x=0}^{20} t^x P(X=x) \\ &= \sum_{x=0}^{20} t^x \binom{20}{x} (0.6)^x (1-0.6)^{(20-x)} \\ &= \sum_{x=0}^{20} \binom{20}{x} (0.6t)^x (1-0.6)^{(20-x)} \\ &= (0.6t + (1-0.6))^{20} \\ &= (0.4 + 0.6t)^{20} \end{aligned}$$

For MGF, replace t by e^t in the above expression

$$M_X(t) = (0.4 + 0.6e^t)^{20} \quad [4]$$

(ii)

Using the fact that $\frac{\frac{S_A^2}{\sigma_A^2}}{\frac{S_B^2}{\sigma_B^2}}$ is $F_{12,14}$ distribution

$$\begin{aligned} P\left(\frac{S_A^2}{S_B^2} < 4.75\right) &= P\left(\frac{\frac{S_A^2}{\sigma_A^2}}{\frac{S_B^2}{\sigma_B^2}} < 4.75 \frac{\sigma_B^2}{\sigma_A^2}\right) = P\left(F_{12,14} < 4.75 \frac{8}{10}\right) \\ &= P(F_{12,14} < 3.8) = 1 - P(F_{12,14} > 3.8) = 1 - 0.01 = 0.99 \quad [3] \end{aligned}$$

(iii)

$$\begin{aligned}
 V(Y) &= E[V(Y|X)] + V[E(Y|X)] \\
 &= E(X+1) + V(2X+3) \\
 &= E(X) + 1 + 4V(X) \\
 &= 5 + 1 + 4(5) \\
 &= 26
 \end{aligned}$$

[3]

[10 Marks]**Solution 3:**

(i)

We can write $P(N = n) = \frac{n+1-1}{(n+1)!} = \frac{1}{n!} - \frac{1}{(n+1)!}$

$F(n) = P(N \leq n)$

$$\begin{aligned}
 &= 0 \quad \text{for } n < 0 \\
 &= \sum_{t=0}^{[n]} \left(\frac{1}{t!} - \frac{1}{(t+1)!} \right) \\
 &= \left(\frac{1}{0!} - \frac{1}{1!} \right) + \left(\frac{1}{1!} - \frac{1}{2!} \right) + \left(\frac{1}{2!} - \frac{1}{3!} \right) + \dots + \left(\frac{1}{[n]!} - \frac{1}{[n+1]!} \right) \\
 &= 1 - \frac{1}{([n+1])!} \quad \text{for } 0 \leq n < \infty, \quad [n] \text{ is integer part of } n
 \end{aligned}$$

[4]

(ii)

The probability generating function of N is $E(t^N)$

$$\begin{aligned}
 G_N(t) &= E(t^N) = t^0 \left(\frac{1}{0!} - \frac{1}{1!} \right) + t^1 \left(\frac{1}{1!} - \frac{1}{2!} \right) + t^2 \left(\frac{1}{2!} - \frac{1}{3!} \right) + \dots \\
 &= \left(1 + t + \frac{t^2}{2!} + \dots \right) - \left(1 + \frac{t}{2!} + \frac{t^2}{3!} + \dots \right) \\
 &= \left(1 + t + \frac{t^2}{2!} + \dots \right) - \frac{1}{t} \left(t + \frac{t^2}{2!} + \frac{t^3}{3!} + \dots \right) \\
 &= e^t - \frac{1}{t} (e^t - 1) \\
 &= e^t \left(1 - \frac{1}{t} \right) + \frac{1}{t} \\
 &= \frac{1 + (t-1)e^t}{t}
 \end{aligned}$$

[4]

(iii)

Differentiating the above PGF w.r.t. to t gives

$$G'(t) = -t^{-2} + e^t - \{e^t(-t^{-2}) + e^t t^{-1}\}$$

Substituting $t=1$

$$\begin{aligned}
 G'(t) &= -1^{-2} + e^1 - \{e^1(-1^{-2}) + e^1 1^{-1}\} \\
 &= -1 + e - (-e + e) \\
 &= e - 1
 \end{aligned}$$

[3]

[11 Marks]

Solution 4:

(i)

P (Critical|Survived) =

$$\frac{P(\text{Survived}|\text{Critical}) \cdot P(\text{Critical})}{P(\text{Survived}|\text{Critical}) P(\text{Critical}) + P(\text{Survived}|\text{Serious}) P(\text{Serious}) + P(\text{Survived}|\text{Stable}) P(\text{Stable})}$$

$$= \frac{0.5 (0.2)}{0.5 (0.2) + 0.8 (0.35) + 0.95 (0.45)} = \frac{0.1}{0.8075} = 0.1238 \quad [4]$$

(ii)

Given:

Let $\bar{X}_{100}$ denote the average pregnancy period in days.

$$\bar{X}_{100} \sim N(268, \frac{16^2}{100}) = N(268, 2.56)$$

We need to find $P(\bar{X}_n > 265)$

$$P\left\{\frac{\bar{X}_n - 268}{\sqrt{2.56}} > \frac{265 - 268}{1.6}\right\} = P(Z > -1.875) = P(Z < 1.875) = 0.9696 \quad [3]$$

[7 Marks]**Solution 5:**

(i)

Let X be the number in the sample who support the economic reforms

 $X \sim \text{Binomial}(500, 0.4)$ $E(X) = 200$; $\text{Var}(X) = 120$

The normal approximation to the Binomial gives, using a continuity correction

$$P(X \geq 220) = P(X \geq 219.5) = P(Z \geq 1.78) = 1 - P(Z \leq 1.78) \\ = 1 - 0.96246 = 0.03754 \quad [3]$$

(ii)

$$P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$$

Here $p = 0.000125$; $n = 50,000$ So $\lambda = np = 50,000 (0.000125) = 6.25$

Using the Poisson Probability Function

$$P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$$

$$= \frac{e^{-6.25} 6.25^0}{0!} + \frac{e^{-6.25} 6.25^1}{1!} + \frac{e^{-6.25} 6.25^2}{2!}$$

$$= 0.0517$$

$$P(X \leq 2) = P\left(Z < \frac{2.5 - 6.25}{\sqrt{6.25}}\right) = P(Z < -1.5) = 1 - P(Z < 1.5) = 1 - 0.93319$$

$$= 0.06681 \quad [3]$$

[6 Marks]

Solution 6:

(i)

Let Y_1, Y_2, Y_3, Y_4 denote the mgs of drug to be observed. We know that the Y_i 's are normally distributed with mean $\mu = 250$ and variance $\sigma^2 = 1$ for $i = 1, 2, 3, 4$.

$$\text{Sample mean } \bar{Y} \sim N\left(\mu, \frac{\sigma^2}{n}\right) = N(\mu, 0.5^2)$$

$$\text{Now we want } P(\mu - 0.2 \leq \bar{Y} \leq \mu + 0.2) = P(-0.2 \leq \bar{Y} - \mu \leq 0.2)$$

$$= P\left(\frac{-0.2}{0.5} \leq \frac{\bar{Y} - \mu}{0.5} \leq \frac{0.2}{0.5}\right) = P(-0.4 \leq Z \leq 0.4)$$

$$= 2P(Z \leq 0.4) - 1 = 2(0.65542) - 1 = 0.31084 \text{ i.e. } 31\% \quad [3]$$

The probability of sample mean lies in the interval (249.8 mg, 250.2 mg) is 31%.

(ii)

$$\text{Now we want } P(\mu - 0.4 \leq \bar{Y} \leq \mu + 0.4) = P(-0.4 \leq \bar{Y} - \mu \leq 0.4) = 0.99$$

Dividing each term of the inequality by standard deviation $\left(\frac{\sigma}{\sqrt{n}}\right)$ and using $\sigma = 1$

$$\text{We get, } P(-0.4\sqrt{n} \leq Z \leq 0.4\sqrt{n}) = 0.99$$

$$\text{From the tables, we know that } P(-2.5758 \leq Z \leq 2.5758) = 0.99$$

$$\text{Hence, } n = \left(\frac{2.5758}{0.4}\right)^2 = 41.4672$$

A sample of size 41 cannot attain our objective. At $n = 42$, the probability of sample mean lie in the interval (249.6 mg, 250.4 mg) slightly exceeds 99%. [3]

[6 Marks]**Solution 7:**

(i)

The likelihood is:

$$\begin{aligned} L(\mu, \sigma^2) &= \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} \frac{1}{x_i} \exp\left\{-\frac{1}{2}\left(\frac{\ln x_i - \mu}{\sigma}\right)^2\right\} \\ &= \text{const} \times \frac{1}{\sigma^n} \times \exp\left\{-\frac{1}{2\sigma^2} \sum_{i=1}^n (\ln x_i - \mu)^2\right\} \end{aligned}$$

Taking logs, we get:

$$\ln L(\mu, \sigma^2) = \text{const} - n \ln \sigma - \frac{1}{2\sigma^2} \sum_{i=1}^n (\ln x_i - \mu)^2$$

Differentiating with respect to μ :

$$\frac{\partial}{\partial \mu} \ln L(\mu, \sigma^2) = \frac{1}{\sigma^2} \sum_{i=1}^n (\ln x_i - \mu)$$

Setting this equal to zero (and multiplying through by σ^2) gives:

$$\sum_{i=1}^n (\ln x_i - \hat{\mu}) = 0 \Rightarrow \sum_{i=1}^n \ln x_i - n\hat{\mu} = 0 \Rightarrow \hat{\mu} = \frac{1}{n} \sum_{i=1}^n \ln x_i$$

Differentiating with respect to σ :

$$\frac{\partial}{\partial \sigma} \ln L(\mu, \sigma^2) = -\frac{n}{\sigma} + \frac{1}{\sigma^3} \sum_{i=1}^n (\ln x_i - \mu)^2$$

Setting this equal to zero (and multiplying through by σ^3) gives:

$$-n\hat{\sigma}^2 + \sum_{i=1}^n (\ln x_i - \hat{\mu})^2 = 0 \Rightarrow \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (\ln x_i - \hat{\mu})^2$$

Now expanding the brackets and then substituting for $\hat{\mu}$ we get:

$$\begin{aligned} \hat{\sigma}^2 &= \frac{1}{n} \sum_{i=1}^n \{ (\ln x_i)^2 - 2\hat{\mu} \ln x_i + \hat{\mu}^2 \} \\ &= \frac{1}{n} \left\{ \sum_{i=1}^n (\ln x_i)^2 - 2\hat{\mu} \sum_{i=1}^n \ln x_i + n\hat{\mu}^2 \right\} \end{aligned} \quad [4]$$

(ii)

From page 23 of the Tables, we have:

$$CRLB = -\frac{1}{E \left[\frac{\partial^2}{\partial \mu^2} \ln L(\mu) \right]}$$

So we need the second log-differential:

$$\frac{\partial^2}{\partial \mu^2} \ln L(\mu) = -\frac{n}{\sigma^2}$$

Now since σ^2 is a constant, we have:

$$E \left[\frac{\partial^2}{\partial \mu^2} \ln L(\mu) \right] = E \left[-\frac{n}{\sigma^2} \right] = -\frac{n}{\sigma^2}$$

Hence, $CRLB = \frac{\sigma^2}{n}$

We know that MLE $\hat{\mu}$ is asymptotically normally distributed i.e.

$$\hat{\mu} \sim N(\mu, CRLB)$$

So a 95% confidence interval for μ is given by:

$$\hat{\mu} \pm 1.96\sqrt{CRLB} = \hat{\mu} \pm 1.96\sqrt{\frac{\sigma^2}{n}} \quad [3]$$

(iii)

Let $Y = \ln(X)$; then $Y \sim N(\mu, \sigma^2)$

From (i) above we have $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n \ln X_i = \frac{1}{n} \sum_{i=1}^n Y_i$

The bias of $\hat{\mu}$ is given by $\text{Bias}(\hat{\mu}) = E(\hat{\mu}) - \mu$

$$E(\hat{\mu}) = E\left(\frac{1}{n} \sum Y_i\right) = \frac{1}{n} \sum E(Y_i) = \frac{1}{n} \sum \mu = \frac{1}{n} n\mu = \mu$$

Hence $\text{bias}(\hat{\mu}) = 0$, so $\hat{\mu}$ is unbiased.

The estimator is consistent if its mean square error (MSE) tends to zero as $n \rightarrow \infty$.

The MSE of $\hat{\mu}$ is given by $\text{MSE}[\hat{\mu}] = \text{Var}[\hat{\mu}] + (\text{Bias}[\hat{\mu}])^2$

Since the bias is zero, the MSE is:

$$\text{MSE}(\hat{\mu}) = \text{var}(\hat{\mu}) = \text{var}\left(\frac{1}{n} \sum Y_i\right) = \frac{1}{n^2} \sum \text{var}(Y_i) = \frac{1}{n^2} \sum \sigma^2 = \frac{\sigma^2}{n}$$

This is consistent as MSE tends to zero for large n . [4]

(iv)

From (ii) and (iii) above $\text{CRLB} = \frac{\sigma^2}{n} = \text{var}(\hat{\mu})$. So $\hat{\mu}$ attains the CRLB. [1]

[12 Marks]

Solution 8:

(i)

Let n be the sample size and p the underlying population proportion who are aware of the shop. The number of people who are aware of the shop, X , is distributed as $X \sim \text{Bin}(n, p) \approx N(np, npq)$

Hence, the estimator of the population proportion, $\hat{p} = \frac{X}{n}$ is distributed as

$$\hat{p} \approx N\left(p, \frac{pq}{n}\right) \Rightarrow \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}} \approx N(0, 1)$$

The asymptotic 90% confidence interval for p is given by

$$\hat{p} \pm 1.6449 \sqrt{\frac{pq}{n}}$$

Since the interval is symmetric about $\hat{p}$, we require $1.6449 \sqrt{\frac{pq}{n}} \leq 0.10$

Further, $pq = p(1-p)$ has a maximum value when $p = \frac{1}{2}$
Hence the confidence interval will be widest when $p = q = 0.5$

So we must choose n so that $1.6449 \sqrt{\frac{(0.5)(0.5)}{n}} \leq 0.10 \Rightarrow n \geq 67.6424$

Therefore minimum sample size required is 68 people. [4]

(ii)

We know that if $X \sim \text{Exp}(\lambda)$ then $\bar{X} \sim \text{Gamma}(n, n\lambda)$

If $Y \sim \text{Gamma}(\alpha, \lambda)$ then $2\lambda Y \sim \chi^2_{2\alpha}$

Therefore, $2n\lambda \bar{X} \sim \chi^2_{2n}$ and we can use the tables of the χ^2 distribution to find a confidence interval.

For given data $18\lambda \bar{X} \sim \chi^2_{18}$ and $\bar{X} = \frac{67}{9}$.

Using the values in the Tables, we have $8.231 < 134\lambda < 31.53$

So, the confidence interval for λ is $(0.0601, 0.2301)$ [4]

[8 Marks]

Solution 9:

(i)

We are interested in the hypothesis that the manager's assumption is incorrect. This can be formally written as $H_1: \mu > 60$, where μ is the mean number of sales contacts per month. Thus, we are interested in testing $H_0: \mu = 60$ against $H_1: \mu > 60$.

For large enough n , the sample mean $\bar{X}$ is a point estimator of μ that is approximately normally distributed $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$. Hence, our test statistic is $Z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$

Rejection region, with $\alpha = 10\%$ is given by $\{z > z_{0.10} = 1.2816\}$ from Tables.

The population variance σ^2 is not known, but it can be estimated very accurately (because $n = 36$ is sufficiently large) by the sample variance $S^2 = 144$.

Thus, the observed value of the test statistic is approximately

$$Z = \frac{\bar{X} - \mu_0}{\frac{S}{\sqrt{n}}} = \frac{68 - 60}{\frac{12}{\sqrt{36}}} = 4$$

Since Z lies in the rejection region (as $z = 4$ exceeds $z_{0.10} = 1.2816$), we reject

$H_0: \mu = 60$. Thus, at $\alpha = 10\%$ level of significance, the evidence is sufficient to indicate that manager's assumption is incorrect and that the average number of sales contacts per month exceeds 60. [4]

(ii) In (i) above rejection region was given by $Z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}} > 1.2816$ which is equivalent to

$$\bar{X} - \mu_0 > 1.2816 \frac{\sigma}{\sqrt{n}} \text{ or } \bar{X} > \mu_0 + 1.2816 \frac{\sigma}{\sqrt{n}}$$

Substituting $\mu_0 = 60$ and $n = 36$ and using S to approximate σ , we find the rejection region to be $\bar{X} > 60 + 1.2816 \frac{12}{\sqrt{36}} \Rightarrow \bar{X} > 62.5632$

Power of the test is probability of rejecting H_0 when it is false.

Power of the test when mean is 64:

$$= P(\bar{X} \geq 62.5632 \text{ when } \mu = 64) = P\left(\frac{\bar{X} - \mu_1}{\frac{\sigma}{\sqrt{n}}} \geq \frac{62.5632 - 64}{\frac{12}{\sqrt{36}}}\right)$$

$$= P(Z \geq -0.72) = P(Z \leq 0.72) = 0.76424. \text{ i.e. } 76.4\%$$

$$\text{Power of the test when mean is 66} = P\left(Z \geq \frac{62.5632 - 66}{\frac{12}{\sqrt{36}}}\right) = P(Z \geq -1.72)$$

$$= P(Z \leq 1.72) = 0.95728. \text{ i.e. } 95.7\% \quad [4]$$

(iii) Power of the test increases as the means in the alternative hypothesis moves away from the mean for the null hypothesis. [1]

[9 Marks]

Solution 10:

Test for association:

H_0 : There is no association between policy size and policy withdrawals

H_1 : There is an association between policy size and policy withdrawals

The observed numbers in each category are:

OBSERVED	Withdrawals	Non-Withdrawals	Total
Small size	450	1050	1500
Large size	100	400	500
Total	550	1450	2000

The expected numbers for each category are:

EXPECTED	Withdrawals	Non-Withdrawals	Total
Small size	412.5	1087.5	1500
Large size	137.5	362.5	500
Total	550	1450	2000

The chi square statistic can then be calculated:

$$\sum \frac{(O - E)^2}{E} = \frac{(450 - 412.5)^2}{412.5} + \frac{(1050 - 1087.5)^2}{1087.5} + \frac{(100 - 137.5)^2}{137.5} + \frac{(400 - 362.5)^2}{362.5}$$

$$= 18.809$$

The number of degrees of freedom is $(2-1) \times (2-1) = 1$

Since the observed value of the test statistic exceeds 6.635, the upper 1% point of the χ^2_1 distribution, we reject the null hypothesis and conclude that there is an association between policy size and policy withdrawals.

[5 Marks]

Solution 11:

(i)

We know that $\hat{\beta} = \frac{S_{xy}}{S_{xx}} = 782 / 2740 = 0.2854$

And $\hat{\alpha} = \bar{Y} - \hat{\beta} \bar{X} = 55 - 0.2854 (159) = 9.6212$

Hence, the regression model is $Y = 9.6212 + 0.2854X$ [2]

(ii)

Sample correlation coefficient, $r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} = \frac{782}{\sqrt{(2740)(438)}} = 0.7138$

From page 25 of the Tables, we have: $z_r \sim N(z_\rho, \frac{1}{n-3})$ where

$$z_r = \tanh^{-1} r = \frac{1}{2} \ln \frac{1+r}{1-r} \quad \text{and} \quad z_\rho = \tanh^{-1} \rho = \frac{1}{2} \ln \frac{1+\rho}{1-\rho}.$$

90% confidence limits for ρ : $\tanh(\tanh^{-1} r \pm 1.6449 \sqrt{\frac{1}{n-3}})$

Substituting values of $r = 0.7138$ and $n = 9$ we get 90% confidence limits for ρ as (0.2198, 0.9165) [3]

(iii) We have $\hat{\sigma}^2 = \frac{1}{n-2} \left(S_{yy} - \frac{(S_{xy})^2}{S_{xx}} \right) = \frac{1}{7} \left(438 - \frac{782^2}{2740} \right) = 30.69$

We know $\frac{7\hat{\sigma}^2}{\sigma^2} \sim \chi^2_7$, which gives a confidence interval for σ^2 as

$$\left(\frac{7(30.69)}{14.07}, \frac{7(30.69)}{2.167} \right)$$

$$= (15.25, 99.13) \quad [3]$$

(iv)

To check the fit of a linear regression model, we can:

- Calculate the proportion of the variation explained by the model (i.e. coefficient of determination)
- Plot the residuals to check that they are normally distributed
- Check the sizes of the residuals are acceptable with the value of estimated standard deviation of the error distribution
- Plot the residuals against x (or y) to check that they are pattern-less (i.e. they have random variation) [4]

[12 Marks]

Solution 12:

(i)

 H_0 : Each method has the same average amount of oil extracted from the Shale H_1 : There are differences among the average amount of oil extracted from the Shale by different methods.

For the given data, summary measures are:

$$y_{1.} = 8; \quad y_{2.} = 11; \quad y_{3.} = 16; \quad y_{..} = 35; \quad \sum \sum y_{ij}^2 = 121$$

$$SS\ T = 121 - 35^2/12 = 18.9167$$

$$SS\ B = (8^2/4 + 11^2/4 + 16^2/4) - 35^2/12 = 8.1667$$

$$SS\ R = 18.9167 - 8.1667 = 10.75$$

Source of variation	Degrees of Freedom	Sum of squares	Mean squares
Between treatments	2	8.1667	4.0834
Residuals	9	10.75	1.1944
Total	11	18.9167	

The variance ratio is $F = 4.0834 / 1.1944 = 3.4186$

Under H_0 , this has an $F(2, 9)$ distribution. The 5% critical point is 4.256, so we do not reject H_0 , and we conclude that the average amount of oil extracted doesn't differ between the three methods.

The assumptions are: The underlying population distribution is normal with common variance. It is also assumed that the samples have been drawn randomly and independently of each other. [6]

(ii) We are testing:

$$H_0: \mu_1 = \mu_3 \text{ vs. } H_1: \mu_1 < \mu_3$$

Under H_0 , the statistic

$$\frac{(\bar{y}_{3.} - \bar{y}_{1.}) - (\mu_3 - \mu_1)}{\hat{\sigma} \left(\frac{1}{n_1} + \frac{1}{n_3} \right)^{0.5}}$$

has t_{n-k} distribution.

$$\text{Sample means: } \bar{y}_{1.} = 2.00, \quad \bar{y}_{3.} = 4.00$$

$$\text{The observed value of test statistic is } \frac{(4-2) - 0}{(1.1944)^{0.5} (1/4 + 1/4)^{0.5}} = 2.58799$$

The 5% critical point for t_9 is 1.833, so we reject H_0 , and conclude that the average amount of oil extracted by method 3 is greater than that of method 1. [3]

[9 Marks]
